gabriela olivera
DATA ANALYST

METHODOLOGY · VERSION 0.9

Data Consolidation and Modelling Methodology

Project Data source consolidation and analytical reporting for a small grain (maize) business

Location Mount Maunganui, Bay of Plenty, New Zealand

Tools Microsoft Excel, Power Query (M), Data Model / Power Pivot (DAX), PivotTables

Date 08/10/2026

Analyst Gabriela Olivera, Data Analyst · hello@gabrielaolivera.nz · 027 277 1269 · gabrielaolivera.nz

Engagement Pro bono, in exchange for a business reference

Version 0.9

The full data dictionaries, findings log and decision log are in the Appendix.

1. Purpose and scope

This document sets out the method used to consolidate several raw data sources (CSV, PDF, Excel and image) into one Excel workbook, build a relational data model and produce a PivotTable-based report.

The aim is a process that is **repeatable, auditable and transferable**: anyone with working Excel knowledge should be able to see where each figure comes from, what was changed, and how to refresh the report with new data.

In scope: data ingestion with Power Query, cleansing and validation, data model design, measures and PivotTable reporting, and documentation of findings, decisions and quality checks.

Out of scope: commercial, accounting or tax advice, and automation outside Excel (external databases, cloud services).

2. Confidentiality and use of tools

PRINCIPLE	HOW IT APPLIES
Client data	Kept only in the local working environment. Not shared with third parties or with AI tools.
Analysis	All analytical work on real data is done manually by the analyst.
Technical support	AI is used only for generic technical support (M and DAX syntax, best practice) and for drafting wording and templates, always with made-up sample data.
This document	Contains no client names, figures or files. Sources use generic codes (F01 to F05).

3. Sources

ID	TYPE	GENERIC DESCRIPTION	STAGING QUERY	STATUS
F01	CSV	Item (product) master list from the accounting system	`dim_product`	Loaded
F02	PDF	Inventory items report (sales by item, no dates)	`fct_sales_by_item`	Loaded
F03	PDF	Annual financial statements (income and expenses schedule, profit and loss)	`fct_income_expenses`	Loaded
F04	Image	Purchases record: date, quantities, supplier and transport type	`fct_pickups`	Loaded (transcribed manually)
F05	Export	Contacts list from the accounting system	n/a	Excluded (see D07)

4. Process architecture

Each transformation has one logical home:

```
[Raw sources] → [Staging] → [Transformation] → [Data Model] → [Report]
 CSV/PDF/XLSX/IMG   stg_      int_ / dim_ / fct_   Relationships + DAX   PivotTables
```

LAYER	PURPOSE	LOAD SETTING
Parameters (`prm_`)	Folder paths and configurable values, so no path is hard-coded in a query.	Connection only
Staging (`stg_`)	One query per source: read, promote headers, set basic types. No business logic.	Connection only
Intermediate (`int_`)	Merges, appends and business rules.	Connection only
Dimensions (`dim_`) and facts (`fct_`)	Descriptive tables and transactional tables.	Data Model
Functions (`fn_`)	Reusable logic, such as standard text cleansing.	n/a

Naming: queries use a layer prefix and a `snake_case` name; columns have descriptive names with the unit where relevant; Power Query steps are renamed to say what they do; DAX measures have readable names.

5. Ingestion and cleansing standards

By source type

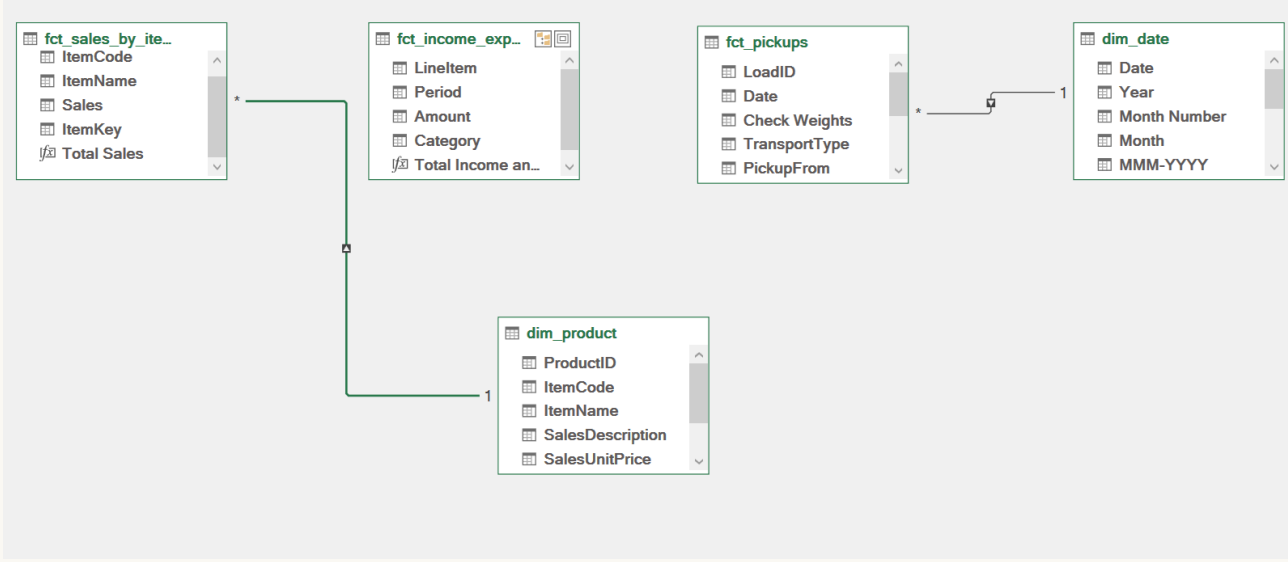
- **CSV:** check delimiter, encoding and locale; confirm decimal separator and date format before setting types.
- **PDF:** check whether the connector finds tables or only pages; review tables split across pages; reconcile extracted totals with the totals printed on the document.
- **Excel:** connect to named tables rather than sheets; watch for merged cells, title rows and embedded subtotals.
- **Image:** Power Query cannot read images. The values are transcribed into an Excel table and checked cell by cell against the image, then ingested as an Excel source.

Cleansing standards

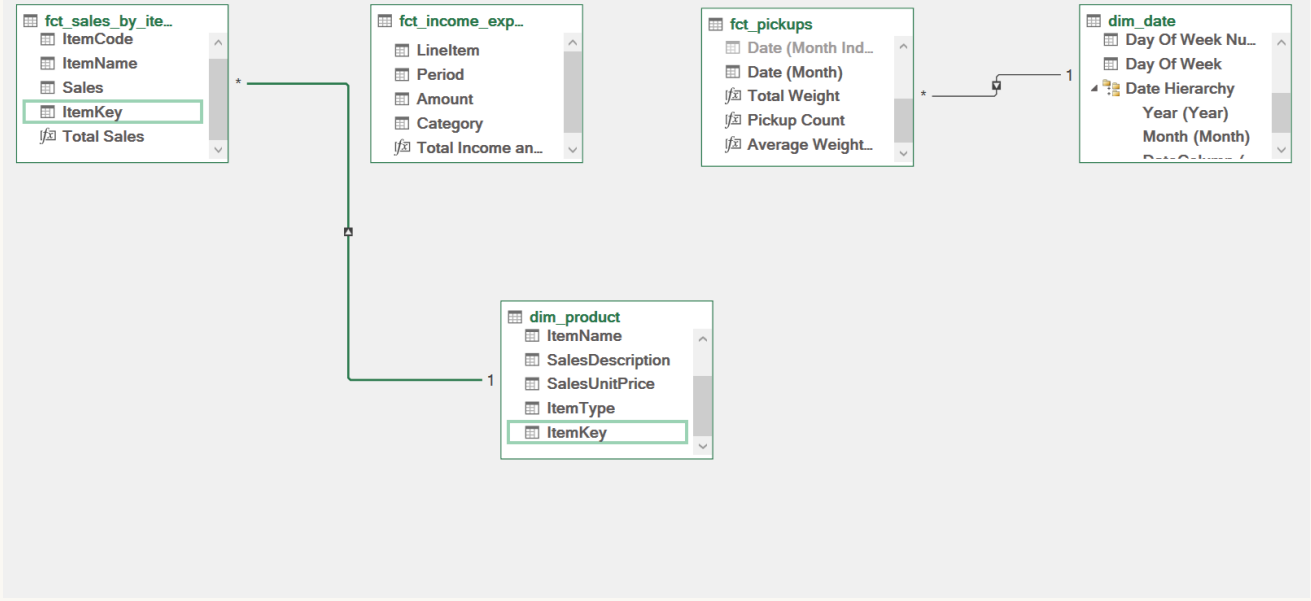
AREA	STANDARD
Data types	Set explicitly in staging, with a check for errors after each change.
Locale	Use "Change Type > Using Locale" when dates or decimals arrive in a different format.
Text	`Trim` , `Clean` and consistent capitalisation on keys and categories.
Units and currency	One unit per measure and one reporting currency (NZD), with GST treatment stated.
Nulls, blanks, duplicates	An explicit rule per column. Define the unique key before removing duplicates.
Keys	Standardise codes across sources, with a mapping where names differ.

6. Data model

Approach: star schema. Fact tables hold keys and numbers; dimension tables describe them, with one unique key per row. Relationships are one-to-many with a single filter direction (dimension to fact), and a continuous `dim_date` calendar table is marked as the date table.



DATA MODEL: TABLES AND FIELDS



DATA MODEL: MEASURES AND DATE HIERARCHY

TABLE	TYPE	RELATED TO	CARDINALITY
dim_product	Dimension	fct_sales_by_item (on ItemKey)	One to many
fct_sales_by_item	Fact (one row per item, no date)	dim_product	Many to one
fct_income_expenses	Fact (one row per line item and period)	None. Periods are handled through the Period column (see D14)	n/a
fct_pickups	Fact (one row per load)	dim_date (on Date)	Many to one

Two design points are worth knowing. Sources are related only through keys that exist in the data, never by guessing (see D03), and item codes are standardised into an ItemKey (spaces removed, upper case) because each source writes them differently (see H18). Column-level definitions and query steps are available on request.

7. Quality checks

CHECK	RESULT
Row counts: rows in each source vs rows loaded	Passed for all four loaded sources
Control totals: sums vs the totals in the original source	Passed for F02 and F03. For F04 the written total differs slightly from the entries (see H06); entries kept as recorded
Orphan keys: facts with no match in their dimension	Passed (sales by item through ItemKey, pickups by date)
Dimension duplicates: key uniqueness	Passed (ItemCode unique across all rows)
Type errors: zero errors in typed columns after refresh	Passed
Full refresh: "Refresh All" completes without errors	Passed

8. Reporting

PivotTables are built on the Data Model, not on worksheet ranges, with explicit DAX measures and slicers connected to the dimension tables. Every metric is defined:

METRIC	DEFINITION	HOW IT IS CALCULATED
Total sales (by period)	Income for the selected period	Total Income and Expenses := SUM([Amount]), with Category = Income and Period filtered
Total expenses	Expenses for the selected period	The same measure, with Category = Expense
Margin	Share of sales left after expenses	(Total sales − Total expenses) ÷ Total sales
Expenses as % of sales	How much of each dollar of sales is spent	Total expenses ÷ Total sales
Total sales (by product)	Sales per item over the whole report period	Total Sales := SUM([Sales])

METRIC	DEFINITION	HOW IT IS CALCULATED
Tonnes picked	Total weight collected	`Total Weight := SUM([Check Weights])`
Average weight per pickup	Average weight of one pickup	`Average Weight per Pickup := AVERAGE([Check Weights])`
Pickup count	Number of pickups in the period	`Pickup Count := COUNTA([LoadID])`

9. Main findings and decisions

The full logs (18 findings and 17 decisions) are available on request. These are the ones that shaped the work most.

REF	WHAT HAPPENED	HOW IT WAS HANDLED
H03	The purchases record does not identify the product	Sources linked only by date and supplier. No product-level analysis is made from it. Recommended to the business: record the type of grain on each pickup
H13	The purchases record is an image, and automatic reading returned many low-confidence cells	Transcribed manually and checked against the image (D12)
H15	The inventory report has period totals only, with no dates	Sales by product are labelled as covering the whole report period. Sales for a period come from the income lines (D14)
H16	Start-up and investment costs are mixed with operating expenses	Period comparisons carry a note. Recommended to the business: a separate category
H18	Item codes differ in spacing and capitalisation between sources	`ItemKey` built in both tables and used for the relationship (D17)

REF	DECISION	WHY
D02	Star schema in the Data Model	Avoids duplication and keeps DAX measures simple
D03	Do not infer missing products	Inferred values cannot be defended or audited
D09	Load only the income and expenses schedule from the annual report	The other tables do not link to products, dates or suppliers, and some hold sensitive details
D14	Period sales from the income lines; product sales from the inventory report	The inventory report has no dates
D17	Relate products on a standardised `ItemKey`, keeping `ItemCode` unchanged	Original codes stay available, and the key is rebuilt the same way on each refresh

10. Glossary

TERM	DEFINITION
Power Query	Excel's tool for connecting to, transforming and combining data (M language).
DAX	Formula language of the Data Model, used for measures.
Fact table	A table of measurable events or transactions (sales, purchases).
Dimension table	A descriptive table that gives context to facts (product, date).
Star schema	A central fact table linked to several dimension tables.